

Computer Programming

Topics

- **Basic elements of C++**
- **C++ Control Statements**
- **Arrays, Strings and Pointers**
- **Modular Programming**
- **User Defined Data Types**
- **File Handling**

Basic elements of C++

- **Variables and Datatypes**

1. What are the values of A, B, C and D after executing the following C++ code fragment?

```
int A = 5, B = 9;  
int C = ++A - B;  
int D = C + B--;  
cout<<A<<B<<C<<D;
```

2. Which arithmetic operator does not work with real (floating-point) numbers?

3. What is the value of *y* in the statement?

```
int y = 20 - 4 % 2 * 8/4 + 10.5;
```

- What is the output of the following code?

```
int a=4, b=2, c=7, d=2;  
cout<<(c == a+b || c == d);
```

- What is the output of this program?

```
int y;  
int x=4;  
cout<<--x;  
x+=10;  
y=x++;  
cout<<"  "<<x<<"\n";  
cout<<y<<"\n";
```

C++ Control Statements

- If this code fragment executed, what would the output be?

```
int a = 3, b = 2, c = 5
if (a > b)
    a = 4;
if ( b > c)
    a = 5;
else
    a = 6;
cout<< a <endl;
```

- What will be the value of **count** and **sum** if a user provides **7.2** as the value of x in the below given C++ code?

```
float sum = 3;
float count = 0;
if (x < 6.0)
    sum = sum + x;
count = (count += 1)* sum;
```

- What is the output of the following C++ code fragment?

```
for (int I = 1; I <= 4; I=I+2)
    cout << I++ << ", ";
cout<<I;
```

- What is the output of the following code?

```
int k = 0;
do {
    cout<< k << "\t";
}while (k < 2);
k = k + 2;
cout<< "\t" << k;
```

- For which values of the integer **x** will the following code become an infinite loop?

```
int number=1;
while (1) {
    cout << number;
    if (number == 3)
        break;
    number += x;
}
```

A. Only 0
C. Only 2

B. Only 1
D. Only 1 or 2

- What will be the value of y, z, and w when x becomes 3

```
int z = 2, w = 1;
for(int x = 2; x <= 3; x++) {
    for(int y = 1; y < 4; y++) {
        if(y <= x)
            z = z + y;
        else
            break;
    }
    w = w + x;
}
```

- Write the output of the following programs .

```
for (int i = 1; i <20;) {
    cout << i++ < ends;
    i = i +2;
}
```

- What is the output of the following code fragment

```
int num=4;
int total=0;
switch (num) {
    case 1:
    case 2: total = 5;
    case 3: total = 10;
    case 4: total = total + 3;
    case 8: total = total + 6;
    default: total = total + 4;
}
cout << total;
```

- What is the output of the following code?

```
char ch='a';  
for(int i=6;i>=1;i--){  
    for(int j=1;j<=(i+1);j++){  
        cout<<ch<<" ";  
        ch++;  
    }  
    cout<<endl;  
}
```

Arrays, Strings and Pointers

- Consider the following statements

```
float Test[8] = {12, 34, 21, 3, 45, 54, 123, 5};  
float *add;
```

One of the following is true

- A. `add = Test[0];`
 - B. `*add = Test;`
 - C. `*(add+3) = &Test[3];`
 - D. `(add+5) = Test[5];`
 - E. None of the above
- Given `int a, int*b = &a;` then one of the following is correct.
 - A. `b = &a`
 - B. `*b = &a`
 - C. `b = a`
 - D. `b = *a`

- What is the output of the following code?

```
char s[ ] = "Addis Ababa University" ;
char t[20] ;
char *ss=s, *tt=t ;
while ( *ss != '\0' )
    *tt++ = *ss++ ;

*tt='\0';
cout<<t;
```

- What is the output of the following code?

```
int numbers[5];
int * p;
p = numbers; *p = 10;
p++; *p = 20;
p = &numbers[2];
*p = 30;
p = numbers + 3;
*p = 40;
p = numbers;
*(p+4) = 50;
for (int n=0; n<5; n++)
    cout << numbers[n] << ", ";
```

- What is the output of the following code?

```
char str [20] = "Tutorial";  
char *p;  
p = str;  
cout<<"The string is " << p<<endl;  
int X = strlen(str);  
cout<<"The value of X is "<<X<<endl;  
strcat (p, " Session");  
cout<<"The final string is "<<str<<endl;
```

- What is the output of the following code?

```
int a[4] = {3, 7, 6, 2};  
int i = 2;  
a[i] = i + 1;  
a[i + 1] = a[ i - 1];  
a[1] = 5;  
for(int j=0;j<4;j++)  
    cout<<a[j]<<" ";
```

Modular Programming

- Consider the following function definition

```
float MyFun (float x, int y) {  
    int Z = x * y;  
    return Z;  
}
```

- Return statements
- Function calling
- Argument passing
- Default parameter
- Function overloading

- What is the output of the following code?

```
int t = 5;
int Mul(int x){
    static int t=2;
    t = ::t * x;
    cout<<"Mul - "<<t<<"\t";
    return t;
}
void main(){
    int t = 3;
    for(int i=0;i<4;i++){
        ::t = Mul(t);
        cout<<"Main-
"<<t<<": "<<::t<<"\n";
    }
}
```

- Consider the following code

```
template<class T, int n>
T maxValue(T value1, T value2) {
    if (value1 > value2)
        return value1;
    else
        return value2;
}
```

- Which one of the below given function calling statements is correct?

- A. `cout<<maxValue <int, char> (1.5, 'A')`
- B. `cout<<maxValue <char, 3> ('A', 'B')`
- C. `cout<<maxValue (1, 2)`
- D. `cout<<maxValue <string> ("AB", "AB")`
- E. `cout<<maxValue <double, double> (1.5, 2.5)`

- What is the output of the following code fragment?

```
int f1(int base, int limit){
    if(base > limit)
        return -1;
    else
        if(base == limit)
            return 1;
        else
            return base * f1(base+1,
                limit);
}
int main() {
    cout << f1(2,4)<<endl;
    return 0;
}
```

User Defined Data Types

- Defining
- Allocating memory space
- Variable or object creation
- Initializing
- Accessing the members

File Handling

- File Streams
- File Variables
- Reading and writing data
 - Text File
 - Binary File
- Passing File Variables to a function
- Sequential and random Access modes

- If `infile.txt` holds 1 2 3 4, what does the following code print?

```
ifstream infile;  
infile.open("infile.txt");  
int num = 0, sum = 0;  
while (infile >> num) {  
    sum += num;  
}  
cout << sum;
```

- What will be written in the text file?

```
fstream pFile;  
char c;  
pFile.open("sample.txt");  
for (c = 'A'; c <= 'E'; c++) {  
    pFile.put(c);  
}  
pFile.close();
```

- What is the output of the following code?

```
ofstream out_file("test10.txt");
    out_file<<"My Computer";
    out_file.close();
    char ch;
    long offset,last;
    ifstream in_file("test10.txt");
    if(in_file.fail()) {
        cout<<"Unable to
open the file\n";
        exit(1);
    }
    in_file.seekg(0L,ios::end);
    last = in_file.tellg();
    offset = 0L;
```

```
for(offset=0L;offset<=last;offset++)
    //while(!in_file.eof()) {

    in_file.seekg(offset,ios::beg);
        ch = in_file.get();
        if(in_file.eof())
            break;
    cout<<ch<<"\n";
        //offset++;
    }
    in_file.close();
```